

Affine-segment preservation on a quarter ball

MathlibAnnex.IsometryEquiv.map_lineMap_of_mem_ball

theorem

Passes from midpoint preservation to every real parameter in a segment.

Statement

Let E, F be real normed spaces, $f : S \rightarrow T$ a bijective isometry between subsets, and $c \in S$. Assume $R > 0$, $B(c, R) \subseteq S$ and $B(f(c), R) \subseteq T$. For $x, y \in S \cap B(c, R/4)$ and $a \in [0, 1]$,

$$f((1 - a)x + ay) = (1 - a)f(x) + af(y).$$

Assumptions

Both radius- R open balls are contained in their respective sets. Both endpoints lie in the source quarter ball. The parameter is restricted to $0 \leq a \leq 1$.

Conclusion

The same isometry preserves every point of the segment between these endpoints, with the same parameter a .

Proof route

On the convex quarter ball, midpoint preservation gives an arithmetic progression at rational subdivision points. Continuity then gives all real parameters in the interval.

▼ Proof steps

1. Put $U = B(c, R/4)$. If $x, y \in U$ and $0 \leq a \leq 1$, then

$$\begin{aligned} \|(1 - a)x + ay - c\| &\leq (1 - a)\|x - c\| + a\|y - c\| \\ &< (1 - a)\frac{R}{4} + a\frac{R}{4} = \frac{R}{4}. \end{aligned}$$

Hence $(1 - a)x + ay \in U \subseteq S$. Restrict f to $g : U \rightarrow F$. By Quarter-ball midpoint theorem, $g((z + w)/2) = (g(z) + g(w))/2$ for all $z, w \in U$.

2. Fix $n \geq 1$ and set

$$z_k = \left(1 - \frac{k}{n}\right)x + \frac{k}{n}y, \quad w_k = g(z_k) \quad (0 \leq k \leq n).$$

Step 1 puts every z_k in U . Since $z_k = (z_{k-1} + z_{k+1})/2$, midpoint preservation gives

$$2w_k = w_{k-1} + w_{k+1}, \quad w_{k+1} - w_k = w_k - w_{k-1}.$$

Thus the successive differences are constant.

3. Summing the constant differences and using $w_0 = g(x)$, $w_n = g(y)$ gives

$$w_k = \left(1 - \frac{k}{n}\right)g(x) + \frac{k}{n}g(y).$$

This proves the desired formula for every rational parameter in $[0, 1]$.

4. For real $a \in [0, 1]$, choose rationals $a_j \rightarrow a$. The isometry is continuous, and

$$\|g((1 - a_j)x + a_j y) - g((1 - a)x + ay)\| = |a_j - a| \|y - x\| \rightarrow 0.$$

The affine combination on the right is continuous as well, so the rational identity passes to the limit. Replacing g by the restriction of f proves the theorem.

Main citations

- Exact theorem and proof
- Quarter-ball midpoint theorem
- Continuous midpoint maps preserve segments

Lean source signature (exact)

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theorem map_lineMap_of_mem_ball
  {s : Set E} {t : Set F} (f : s ≈i t) (c : s) {R : ℝ} (hR : 0 < R)
  (hsball : ball (c : E) R ⊆ s)
  (htball : ball ((f c : t) : F) R ⊆ t)
  (x y : s) (hx : (x : E) ∈ ball (c : E) (R / 4))
  (hy : (y : E) ∈ ball (c : E) (R / 4))
  (a : ℝ) (ha : a ∈ Icc (0 : ℝ) 1) :
  ((f <AffineMap.lineMap (x : E) (y : E) a,
    hsball (ball_subset_ball (by linarith : R / 4 ≤ R)
      ((convex_ball (c : E) (R / 4)).lineMap_mem hx hy ha))) : t) : F) =
  AffineMap.lineMap ((f x : t) : F) ((f y : t) : F) a
```

In the source	Mathematical meaning
$(f : s \approx_i t) (c : s)$	The distance-preserving bijection $f : S \rightarrow T$ and center $c \in S$, with $s \subseteq E, t \subseteq F$ in real normed spaces.
$\{R : \mathbb{R}\} (hR : 0 < R)$	The common radius $R > 0$.
$(hsball : \text{ball } (c : E) R \subseteq s)$	The open ball $B(c, R)$ lies in the source set.
$(htball : \text{ball } ((f c : t) : F) R \subseteq t)$	The open ball $B(f(c), R)$ lies in the target set; this is a separate hypothesis.

In the source	Mathematical meaning
$(x\ y : s)$	Two points $x, y \in S$.
$(hx : (x : E) \in \text{ball } (c : E) (R / 4))$	The inequality $\ x - c\ < R/4$.
$(hy : (y : E) \in \text{ball } (c : E) (R / 4))$	The inequality $\ y - c\ < R/4$ for the other point.
$(a : \mathbb{R}) (ha : a \in \text{Icc } (0 : \mathbb{R}) 1)$	A real coefficient a with $0 \leq a \leq 1$.
$\text{AffineMap.lineMap } (x : E) (y : E) a$	The point $(1 - a)x + ay$ of E .
$(\text{convex_ball } (c : E) (R / 4)).\text{lineMap_mem } hx\ hy\ ha$	The derived membership of that point in the quarter ball, using convexity, both endpoint bounds and $a \in [0, 1]$.
$\text{hsball } (\text{ball_subset_ball}$	The subsequent inclusion into $B(c, R)$ and then S , which makes $f((1 - a)x + ay)$ a valid application.
$\text{AffineMap.lineMap } ((f\ x : t) : F) ((f\ y : t) : F) a$	The output expression $(1 - a)f(x) + af(y)$; the whole conclusion equates it with the image of the source segment point.

Exact source and proof.

Earlier published Card and PDF

▼ Exact content identity

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